

DEVICE / SESSION TAXONOMY AUDIT

PART 3 – FINAL DEVICE / SESSION TAXONOMY

FAMILY 1: AI-GENERATED DEVICE FINGERPRINT SPOOFING

A. Identity

Field	Value
Attack ID	DFS-001
Attack Family	AI-Generated Device Fingerprint Spoofing
Attack Subtypes / Variants	Anti-detect browser automation; Kernel-level fingerprint spoofing; AI-native anti-detection tools; Data wiping device reset
Primary Lifecycle Stage	Device / Session (Stage 3)
Secondary / Cross-stage Stages	Account Creation (Stage 2 – onboarding), Authentication (Stage 4 – login risk)
Target / Actor	Device fingerprinting systems, device reputation engines, risk-based authentication
Attack Objective	Present a synthetic device fingerprint indistinguishable from a legitimate device to evade device-based fraud detection and enable multi-accounting, account creation, and fraud at scale

B. Fraud Understanding

What is the attack?

The attacker uses GenAI-generated device fingerprints to make a single physical device or emulator appear as thousands of unique, legitimate devices. AI-generated identities produce "flawless device fingerprints, clean session behavior, and perfectly consistent form inputs"—the exact things older fraud systems rely on to score 'low risk'. Unlike traditional anti-detect browsers that use template-based fingerprints with detectable artifacts, GenAI enables **indistinguishable** synthetic device profiles.

Traditional mechanism:

Before GenAI, attackers used anti-detect browsers (Multilogin, GoLogin, Incogniton) to spoof canvas, WebGL, audio, fonts, and timezone. Each fingerprint required manual configuration or template selection. Detection was possible through:

- Artifact patterns in canvas/WebGL rendering
- Inconsistent attribute combinations
- Behavioral patterns (no mouse movements, superhuman speed)

GenAI transformation:

GenAI transforms fingerprint spoofing from **template-based, artifact-prone** to **indistinguishable, AI-generated**:

Dimension	Pre-GenAI	GenAI-Transformed
Fingerprint quality	Template-based, detectable artifacts	AI-generated profiles with "flawless" fingerprints, "perfectly consistent" attributes
Scalability	Manual configuration per device	AI agents "spin up fake users at scale"
Automation	Human-operated	"Anti-detection browser server for AI agents" enables autonomous operation
Adaptation	Static templates	Real-time optimization based on detection feedback

Why GenAI is load-bearing:

If GenAI is removed, the attacker loses the ability to:

- Generate **flawless, indistinguishable** fingerprints without detectable artifacts
- Operate at **scale** with consistent quality
- **Automate** fingerprint generation and adaptation
- **Iterate** based on detection feedback

The attack reverts to template-based spoofing that is **detectable** by modern ML-based device fingerprinting systems.

HARD GenAI Removal Test result: PASS — GenAI creates a qualitatively new capability. The transformation from artifact-prone, template-based spoofing to indistinguishable, AI-generated fingerprints is a qualitative leap.

Prerequisites:

- Access to GenAI fingerprint generation tools (commercial: RoxyBrowser, Camoufox; research: Darwinium Beagle)
- Understanding of device fingerprinting signals
- (For scale) AI agent infrastructure for autonomous operation

Attack Flow:

1. Attacker obtains GenAI fingerprint generation capability
2. System generates a synthetic device profile (canvas, WebGL, audio, fonts, timezone, screen resolution, OS, browser)
3. Profile is applied via anti-detect browser, kernel-level spoofing, or AI agent
4. Attacker accesses financial platform with spoofed device fingerprint
5. Platform's device fingerprinting system scores the device as "low risk" (looks identical to legitimate device)
6. Attacker conducts fraud (account opening, transaction, etc.) while evading device-based detection
7. Fingerprint is cycled or regenerated for next session

Downstream payment consequence:

- Mass synthetic account creation without device-based detection
- Bonus/referral abuse at industrial scale
- Evasion of device reputation systems
- Multi-accounting from limited hardware
- Account farming for resale
- Enables all downstream fraud (money laundering, bust-out, mule activity)

C. Detection Intelligence

Observable Signals:

Signal Category	Specific Signals	Detection Method	Cross-account Needed?
Device	Canvas/WebGL rendering artifacts	ML anomaly detection	No
Device	Audio fingerprint consistency	ML anomaly detection	No
Device	Font set consistency	ML anomaly detection	No
Device	Timezone/IP consistency	Rule-based + ML	No
Device	OS/browser attribute consistency	Rule-based + ML	No
Device	Hardware attribute consistency (GPU, RAM)	ML anomaly detection	No
Device	Screen resolution consistency	Rule-based	No
Device	WebGL parameter consistency	ML anomaly detection	No
Session	Multiple accounts from same underlying fingerprint	Graph analytics	Yes
Session	Device fingerprint cycling frequency	Velocity rules	Yes
Network	Multiple accounts sharing device fingerprint components	Graph Neural Network	Yes
Network	Fingerprint patterns correlated across accounts	ML clustering	Yes
Temporal	Rapid device fingerprint changes	Velocity rules	Yes

Signal Category	Specific Signals	Detection Method	Cross-account Needed?
Temporal	Unusual fingerprint generation timing	Rule-based	No

Evasion Considerations:

An adaptive attacker could:

- Use GenAI to generate **unique, non-repeating** fingerprints with no detectable artifacts
- Introduce **realistic variance** in fingerprint attributes
- Use **legitimate device profiles** as training data for fingerprint generation
- **Cycle fingerprints** unpredictably to evade pattern detection
- Use **AI agents** to autonomously optimize evasion based on detection feedback

Legitimate Behavior That Could Look Similar:

- Users with privacy browsers (Brave, Tor) that block fingerprinting signals
- Users with multiple devices (phone, tablet, laptop, desktop) creating different fingerprints
- Users with browser updates changing fingerprint attributes
- Users with VPNs changing IP/geolocation
- Corporate networks with shared IP addresses

False-Positive Risks:

- Legitimate users with privacy-focused browsers may have limited or inconsistent fingerprint signals
- Users who frequently switch devices may appear to have "new" fingerprints
- Browser updates may change fingerprint attributes
- VPN usage may cause geolocation inconsistencies

D. Simulation Intelligence

Field	Value
Simulation Type	Hybrid (Algorithmic + Agentic)
Generation Parameters	Canvas rendering quirks; WebGL capabilities; Audio stack behavior; Installed fonts; Timezone; Screen resolution; Color depth; Pixel ratio; OS version; Browser version; Hardware attributes (GPU, RAM); Device model; Fingerprint quality levels (pristine → realistic noise)
Synthetic Entities	Synthetic device profiles (complete fingerprint bundles); Browser/session environments; Device history (past interactions, reputation-building); IP/network context
Synthetic Device Attributes	Canvas fingerprint (rendering output); WebGL fingerprint (capabilities, rendering); Audio fingerprint (audio stack output); Font list (installed fonts); Screen resolution; Color depth; Pixel ratio; Timezone; Language settings; OS type and version; Browser type and version; Hardware attributes; Device model
Synthetic Session Events	Session establishment with spoofed fingerprint; Cookie/token generation; Session continuity patterns; Session termination/reset
Synthetic Behavioral Events	Mouse movements (AI-generated, human-like); Keystroke patterns (timing, rhythm); Navigation patterns; Form-filling timing; Hesitation/backtracking
Temporal Behavior	Session duration distributions; Time of day patterns; Time between sessions; Fingerprint cycling patterns (for evasion variants)
Network Relationships	Multiple accounts sharing underlying device; Fingerprint pattern clustering; Account-to-account similarity scores
Attack Labels	Spoofing attempt; Fingerprint quality score; Evasion confidence
Variant / Mutation Parameters	Fingerprint quality (pristine → realistic noise); Artifact injection (for evading artifact detection); Fingerprint cycling frequency; Tool variants (anti-detect browser, kernel-level, AI agent)
Legitimate Counterpart	Privacy-focused users; Users with multiple devices; Browser update scenarios; VPN users; Corporate network users

Field	Value
Realistic Edge Cases	Limited fingerprint data (privacy browsers); Inconsistent fingerprints (VPN, location changes); Device upgrades (new OS, browser); Shared devices (family tablet)

E. Evidence

Claim	Source	Publisher	Date	Evidence Type	Confidence
AI-generated identities produce flawless device fingerprints, clean session behavior, and perfectly consistent form inputs	FUGU: "AI identity fraud is the new GenAI frontier"	FUGU (fraud intelligence)	2025	Industry report	VERIFIED
Anti-detection browser server for AI agents bridges AI logic and web execution	RoxyBrowser	RoxyBrowser	2025	Product documentation	VERIFIED
Camoufox spoofs real-device fingerprints at kernel level, driven by Playwright API	Camoufox	Camoufox	2025	Product documentation	VERIFIED
Darwinium's Beagle emulates synthetic identities with spoofed devices, geolocation variance, and behaviorally valid interaction patterns	Darwinium Beagle	Darwinium	2025	Product documentation	VERIFIED
Device fingerprinting is probabilistic and spoofable at scale	Ideem	Ideem	2024	Industry analysis	VERIFIED
Traditional behavioral biometrics solutions struggle to keep up with modern fraud techniques	Oscilar	Oscilar	2025	Industry analysis	REASONABLE INFERENCE

F. Product Mapping

Component	Implementation
Generate	Algorithmic generation of synthetic device profiles with realistic fingerprint attributes; AI-agent-driven browser automation (Playwright/CDP) with spoofed fingerprints; Varied fingerprint quality levels for realism
Defend	ML anomaly detection for fingerprint artifacts; Cross-account device fingerprint correlation (Graph Neural Network); Behavioral consistency verification
Detection Approach	CNN-based artifact detection; Graph Neural Network for fingerprint correlation; Clustering for fingerprint pattern detection
Mitigation	Step-up authentication for unrecognized devices; Enhanced monitoring for multi-account patterns; Blocking for confirmed spoofing
Evaluation	Precision (avoiding false positives on privacy browsers); Recall (catching spoofed fingerprints); AUC; False Positive Rate (legitimate vs. spoofed)

FAMILY 2: AI-DRIVEN EMULATOR FARM FRAUD

A. Identity

Field	Value
Attack ID	EFF-001
Attack Family	AI-Driven Emulator Farm Fraud
Attack Subtypes / Variants	Virtual emulator farms; Cloud phone farms; Physical device farms (merged variant); AI-orchestrated device farms
Primary Lifecycle Stage	Device / Session (Stage 3)
Secondary / Cross-stage Stages	Account Creation (Stage 2 – mass account opening), Payment Initiation (Stage 5), Settlement (Stage 13), Cash-out / Mule (Stage 16)
Target / Actor	Device fingerprinting, device reputation, velocity rules, bot detection
Attack Objective	Operate thousands of synthetic device identities simultaneously to create accounts at scale, harvest incentives, and enable downstream fraud, using AI-driven evasion to avoid detection

B. Fraud Understanding

What is the attack?

Attackers deploy **networks of virtual devices** (emulators) or physical phones to simulate thousands of "unique" users from a single infrastructure. Emulator farms can "spin up fake users at scale" using "AI-driven evasion techniques — device spoofing, emulators, and automated tools that cycle through synthetic device identities faster than legacy detection logic can flag them." This enables mass account creation and fraud at scale.

Traditional mechanism:

Before GenAI, attackers deployed networks of virtual Android instances (emulators) to simulate thousands of "unique" users. Each emulator was manually configured with unique fingerprints. Detection relied on:

- Emulator flags (virtualized environment indicators)
- VM/environment indicators
- Behavioral patterns (automated, consistent)
- Network-level correlation (same IP ranges)

GenAI transformation:

GenAI transforms emulator farms from **detectable, static infrastructure** to **adaptive, evasion-optimized systems**:

Dimension	Pre-GenAI	GenAI-Transformed
Configuration	Manual fingerprint per emulator	AI agents "spin up fake users at scale"
Evasion	Static, detectable patterns	"AI-driven evasion techniques... cycle through synthetic device identities faster than legacy detection can flag"
Scale	Hundreds of emulators	"100,000 injection-style fraud attempts per month, linked to emulator farms"
Behavior	Detectable automation	AI agents mimic human behavior patterns

Why GenAI is load-bearing:

If GenAI is removed, the attacker loses the ability to:

- **Cycle through identities** faster than detection systems can flag them
- **Generate indistinguishable** device fingerprints for each emulator
- **Mimic human behavior** at scale
- **Adapt** evasion strategies based on detection feedback

The attack reverts to static emulator farms that are **detectable** by modern fraud prevention systems.

HARD GenAI Removal Test result: PASS — GenAI creates a qualitatively new capability. The transformation from detectable static infrastructure to adaptive, evasion-optimized systems is a qualitative leap.

Prerequisites:

- Access to emulation infrastructure (Android emulators, cloud phone services)
- GenAI fingerprint generation and evasion optimization
- (For AI-driven) Agent infrastructure for autonomous operation
- Account creation automation

Attack Flow:

1. Attacker sets up emulator farm (virtual or physical)
2. GenAI generates unique, indistinguishable device fingerprints for each emulator
3. AI agents control each emulator, mimicking human behavior
4. Accounts are created at scale across the farm
5. AI adapts fingerprints and behavior based on detection feedback
6. Accounts are used for fraud or sold on account marketplaces
7. Emulators are reset and reused for new campaigns

Downstream payment consequence:

- Mass synthetic account creation at industrial scale
- Bonus/incentive abuse (harvesting sign-up bonuses, referrals)
- Dropper accounts for money laundering and APP scams
- Account farms selling verified accounts
- "Evil emulator farms" have stolen millions from US and EU banks
- Foundation for all downstream payment fraud

C. Detection Intelligence

Observable Signals:

Signal Category	Specific Signals	Detection Method	Cross-account Needed?
Device	Emulator detection (virtualized environment flags)	Rule-based + ML	No
Device	VM/environment indicators	Rule-based + ML	No
Device	Identical or highly similar device fingerprints across accounts	ML anomaly detection	Yes
Device	Unnatural consistency in device attributes	ML clustering	Yes
Device	Cloud/VPS IP ranges	Rule-based	Yes
Session	Rapid account creation velocity	Velocity rules	Yes
Session	Multiple accounts from same IP/device farm	Graph analytics	Yes
Behavioral	Automated, consistent behavior patterns	ML anomaly detection	Yes
Behavioral	Superhuman timing/speed	Rule-based + ML	No
Network	Account-to-account similarity scores	Graph Neural Network	Yes
Network	IP/network correlation	Graph analytics	Yes
Network	Physical location correlation (for physical device farms)	Geo-location + ML	Yes
Temporal	Unusual account creation timing (off-hours)	Rule-based	Yes
Temporal	Identical account creation patterns	ML clustering	Yes

Evasion Considerations:

An adaptive attacker could:

- Use GenAI to **cycle through synthetic device identities** faster than detection can flag
- Use AI agents to **mimic human behavior** at scale
- **Distribute** emulators across different IP ranges and geolocations
- Use **residential proxies** to mask infrastructure
- **Vary** account creation timing and patterns
- Use **physical device farms** to evade emulator detection

Legitimate Behavior That Could Look Similar:

- Legitimate businesses creating multiple accounts (e.g., subsidiaries)
- Corporate networks with shared IP addresses
- Users with multiple devices
- High-volume account creation for legitimate purposes (e.g., promotions)

False-Positive Risks:

- Legitimate high-volume account creation (e.g., business onboarding)
- Corporate network users with shared IPs
- Users with multiple legitimate accounts (joint accounts, business accounts)

D. Simulation Intelligence

Field	Value
Simulation Type	Hybrid (Algorithmic + Agentic)
Generation Parameters	Emulator configuration parameters; Fingerprint generation parameters; Farm size (number of devices); Account creation velocity; Behavioral patterns; Evasion optimization parameters; Physical vs. virtual distribution
Synthetic Entities	Synthetic device profiles (complete fingerprint bundles); Emulator instances; Account creation attempts; Behavioral patterns; Network/identity correlations
Synthetic Device Attributes	Same as DS-01 + Emulator-specific indicators (VM flags, environment signals); Physical device variants (for physical farms)
Synthetic Session Events	Multiple session establishment patterns; Session continuity across accounts; Account creation sequences
Synthetic Behavioral Events	AI-generated human-like behavior (mouse movements, keystrokes, navigation); Evasion pattern optimization; Behavioral consistency across accounts
Temporal Behavior	Account creation timing distributions; Session timing patterns; Evasion optimization timing; Dormancy patterns
Network Relationships	Multiple accounts per device; Multiple devices per IP; Physical location clustering; Account-to-account similarity scores; Fraud ring clustering
Attack Labels	Farm type (emulator/cloud/physical); Farm size; Evasion level; Account creation velocity
Variant / Mutation Parameters	Farm type (emulator vs. physical); Farm size; Evasion sophistication; Behavioral mimicry quality; Account creation velocity
Legitimate Counterpart	Legitimate businesses with multiple accounts; Corporate networks; Users with multiple legitimate accounts
Realistic Edge Cases	Mixed legitimate/fraud accounts; Accounts opened for legitimate reasons then used for fraud; Farm transitions (emulator → physical to evade detection)

E. Evidence

Claim	Source	Publisher	Date	Evidence Type	Confidence
100,000 injection-style fraud attempts per month linked to emulator farms	Andy Hess (LinkedIn post)	Andy Hess (security researcher)	2025	Industry report	VERIFIED
Organized crime rings use virtual emulators and root-level device hacks to hijack application APIs	Biometric Update	Biometric Update	2026	News report	VERIFIED
Criminals deploying AI-driven evasion techniques – device spoofing, emulators – cycle through identities faster than legacy detection can flag	Industry report	Industry report	2025	Industry report	VERIFIED
Emulator farms, AI-driven fraud, and reverse-engineered onboarding flows	Sumsub	Sumsub	2025	Industry report	VERIFIED
Major mobile banking fraud operation using emulated devices stole millions from US/EU banks	IBM Trusteer	IBM	2023	Security research	VERIFIED
Cloud phones used to create dropper accounts for APP scams	Group-IB	Group-IB	2024	Security research	VERIFIED
~2,000-strong device emulator farm in London harvesting bonuses	GeoComply	GeoComply	2024	Industry report	VERIFIED

F. Product Mapping

Component	Implementation
Generate	Algorithmic generation of emulator farms (configuration, fingerprints, network patterns); AI-agent-driven farm orchestration; Behavioral simulation at scale

Component	Implementation
Defend	Emulator/VM detection; Device fingerprint correlation (Graph Neural Network); Velocity anomaly detection; Network-level correlation
Detection Approach	ML classification for emulator detection; Graph Neural Network for device correlation; Anomaly detection for velocity/behavior; Network analytics for IP/geolocation correlation
Mitigation	Block emulator traffic; Step-up authentication for suspicious device farms; Enhanced monitoring for mass account creation; Consortium intelligence sharing
Evaluation	Precision (avoiding false positives on legitimate high-volume accounts); Recall (catching emulator farms); AUC; False Positive Rate

FAMILY 3: AI-POWERED REMOTE ACCESS TROJAN (RAT) / ON-DEVICE FRAUD

A. Identity

Field	Value
Attack ID	RAT-001
Attack Family	AI-Powered Remote Access Trojan (RAT) / On-Device Fraud
Attack Subtypes / Variants	Automated Transfer System (ATS) RATs; Behavioral mimicry RATs; AI-distributed RATs; NFC relay with RAT (hybrid)
Primary Lifecycle Stage	Device / Session (Stage 3)
Secondary / Cross-stage Stages	Authentication (Stage 4), Payment Initiation (Stage 5), AML/Compliance (Stage 17)
Target / Actor	Trusted device context, session integrity, behavioral biometrics, transaction authorization
Attack Objective	Compromise a legitimate user's device, then autonomously execute fraudulent transactions from within the trusted device context while mimicking the user's behavior to evade detection

B. Fraud Understanding

What is the attack?

Attackers infect victim devices with Remote Access Trojans (RATs) that give them live remote control. Modern GenAI-enhanced RATs go beyond simple remote control: they incorporate **Automated Transfer Systems (ATS)** that autonomously execute transactions without human intervention, and **behavioral mimicry** that evades behavioral biometrics.

BioCatch's mid-market customers reported a **55% increase in fraud involving remote access Trojans in 2025**, accounting for 15% of total fraud. ThreatFabric documented "RatOn" which "merges traditional overlay attacks with automatic money transfers" and "seamlessly combines remote access capabilities with NFC relay technology and Automated Transfer System (ATS) functions."

Traditional mechanism:

Before GenAI, attackers used RATs to remotely control victim devices—viewing screens, stealing credentials, and manually executing fraud. Detection was possible through:

- Behavioral biometrics (attacker's interaction patterns differ from victim's)
- Unusual navigation patterns
- Suspicious app installations
- Screen recording/remote access indicators

GenAI transformation:

GenAI transforms RAT-based fraud from **human-controlled** to **autonomous**:

Dimension	Pre-GenAI	GenAI-Transformed
Transaction execution	Manual attacker control	Automated Transfer System (ATS) —RATs autonomously execute transfers

Dimension	Pre-GenAI	GenAI-Transformed
Behavioral mimicry	Limited	AI can mimic human interaction patterns to evade behavioral biometrics
Distribution	Generic	AI-generated phishing/social engineering to distribute RATs
Scale	Manual operation	Autonomous, scalable fraud campaigns

Why GenAI is load-bearing:

If GenAI is removed, the attacker loses the ability to:

- **Execute transactions autonomously** (ATS) without human intervention
- **Mimic human behavior** at the level needed to evade behavioral biometrics
- **Distribute RATs at scale** via AI-generated personalized phishing
- **Adapt** behavior based on detection feedback

The attack reverts to human-controlled RATs that are **detectable** by behavioral biometrics and session analysis.

HARD GenAI Removal Test result: PASS — GenAI creates a qualitatively new capability. The transformation from human-controlled to autonomous, behavior-mimicking RATs is a qualitative leap.

Prerequisites:

- Access to RAT malware (Malware-as-a-Service available, e.g., RatOn, Maverick)
- Knowledge of target banking apps
- (For distribution) Social engineering or phishing capability
- (For ATS) Knowledge of transaction flows and authorization mechanisms

Attack Flow:

1. Victim downloads RAT (via social engineering, phishing, or fake app)
2. RAT establishes remote access and surveillance
3. Attacker (or AI) observes victim's banking patterns
4. AI learns victim's behavioral patterns (typing, navigation, timing)
5. RAT autonomously initiates fraudulent transactions (ATS)
6. AI mimics victim's behavior during transaction to evade detection
7. Funds are transferred to mule accounts
8. RAT may also perform NFC relay for contactless payment fraud

Downstream payment consequence:

- On-device fraud (ODF)—transactions from victim's trusted device
- Credential theft and device surveillance
- Ransomware alongside banking fraud
- NFC relay fraud
- Albiriox, Klopatra, deVixor, Remo, Hook, RatOn are active Android banking RAT families
- ATS enables **autonomous** fraud without human intervention

C. Detection Intelligence

Observable Signals:

Signal Category	Specific Signals	Detection Method	Cross-account Needed?
Device	Device fingerprint (consistent—RAT uses victim's device)	N/A (no detection)	No
Device	Suspicious app installations	Rule-based	No
Device	Screen recording/remote access indicators	ML anomaly detection	No
Session	Session anomalies (pattern changes)	ML anomaly detection	No
Behavioral	Behavioral biometrics anomalies (attacker vs. victim patterns)	ML (behavioral biometrics)	No
Behavioral	Unusual navigation patterns (searching for transfer functions)	ML anomaly detection	No

Signal Category	Specific Signals	Detection Method	Cross-account Needed?
Behavioral	Timing anomalies (too fast, too slow)	ML anomaly detection	No
Behavioral	ATS execution patterns (autonomous, consistent)	ML anomaly detection	No
Transaction	Unusual transaction amounts/beneficiaries	ML anomaly detection	No
Transaction	Rapid transaction sequences	Velocity rules	No
Network	Beneficiary connected to mule networks	Graph analytics	Yes

Evasion Considerations:

An adaptive attacker could:

- Use AI to **mimic victim's behavioral patterns** (typing rhythm, navigation, timing)
- Use ATS to execute transactions at **human-like speeds**
- Use **legitimate victim devices** (no device fingerprint anomalies)
- **Distribute** RATs via AI-generated personalized phishing
- Use **NFC relay** for contactless payment fraud
- **Vary** transaction patterns to avoid detection

Legitimate Behavior That Could Look Similar:

- Legitimate transactions from the victim's device
- Legitimate navigation patterns (normal user behavior)
- Legitimate app installations

False-Positive Risks:

- Behavioral biometrics false positives on legitimate users
- Users with variable behavior (new device, new app)
- Accessibility tools causing unusual behavior patterns

D. Simulation Intelligence

Field	Value
Simulation Type	Agentic (primary) + Algorithmic (transaction component)
Generation Parameters	RAT behavior parameters; ATS transaction sequences; Behavioral mimicry quality; Distribution methods; Device compromise sequences
Synthetic Entities	Victim personas (legitimate users with transaction histories); RAT-infected devices; ATS execution sequences; Behavioral profiles (victim baseline + attacker mimicry)
Synthetic Device Attributes	Device fingerprint (matching victim's actual device); RAT installation indicators; Screen recording/remote access indicators
Synthetic Session Events	Normal session establishment; Session takeover by RAT; ATS session patterns; Session continuity
Synthetic Behavioral Events	Victim's normal behavioral profile (baseline); Attacker behavioral mimicry; ATS execution behavior; Navigation patterns (searching for transfer functions)
Synthetic Transactions	Normal victim transactions (baseline); ATS-executed fraudulent transactions; Beneficiary additions; Funds movement to mule accounts
Temporal Behavior	Transaction timing patterns; Behavioral pattern timing; ATS execution speed; Time between infection and fraud
Network Relationships	Mule network connections; Funds layering path
Attack Labels	RAT type; ATS sophistication; Behavioral mimicry quality; Distribution method
Variant / Mutation Parameters	RAT variant (ATS, NFC relay, overlay); Behavioral mimicry quality; Transaction execution speed; Distribution method
Legitimate Counterpart	Legitimate victim transactions; Normal app installations; Normal behavior patterns

Field	Value
Realistic Edge Cases	Partial behavioral mimicry; RAT detection/removal; Multiple RAT families

E. Evidence

Claim	Source	Publisher	Date	Evidence Type	Confidence
55% increase in fraud involving remote access Trojans in 2025, accounting for 15% of total fraud	BioCatch	BioCatch	2025	Industry report	VERIFIED
RatOn merges traditional overlay attacks with automatic money transfers	ThreatFabric	ThreatFabric	2025	Security research	VERIFIED
RatOn seamlessly combines remote access capabilities with NFC relay and ATS	HEAL Security	HEAL Security	2025	Security research	VERIFIED
Maverick banking trojan employs AI-assisted code generation	Kaspersky	Kaspersky	2025	Security research	VERIFIED
Albiriox enables remote control of infected Android devices for financial fraud	Cleafy	Cleafy	2024	Security research	VERIFIED
SpyNote RAT gives remote control and enables NFC card relay fraud	Group-IB	Group-IB	2024	Security research	VERIFIED
RATs enable remote control of devices, bypassing traditional authentication	BioCatch	BioCatch	2025	Industry report	VERIFIED

F. Product Mapping

Component	Implementation
Generate	Agentic generation of RAT behavior (infection, surveillance, ATS execution); Algorithmic generation of transaction sequences; Behavioral mimicry simulation
Defend	Behavioral biometrics (anomaly detection); Session anomaly detection; Transaction anomaly detection; Device integrity checks
Detection Approach	ML anomaly detection for behavioral patterns; ML classifier for transaction anomalies; Graph Neural Network for beneficiary risk scoring
Mitigation	Block suspicious transactions; Step-up authentication for anomalous behavior; Enhanced monitoring for RAT indicators; Customer notification
Evaluation	Precision (avoiding false positives on legitimate behavior); Recall (catching RAT-executed transactions); AUC; False Positive Rate; Speed (real-time detection)

FAMILY 4: AI AGENT-POWERED AUTOMATED BOT INTERACTION

A. Identity

Field	Value
Attack ID	BOT-001
Attack Family	AI Agent-Powered Automated Bot Interaction
Attack Subtypes / Variants	Human-mimicking credential stuffing; Human-mimicking card testing; Human-mimicking account creation; Autonomous agent fraud
Primary Lifecycle Stage	Device / Session (Stage 3)
Secondary / Cross-stage Stages	Authentication (Stage 4), Account Creation (Stage 2), Payment Initiation (Stage 5)
Target / Actor	Bot detection, behavioral biometrics, velocity rules, fraud scoring
Attack Objective	Operate automated bots that appear indistinguishable from legitimate human users, enabling large-scale credential testing, card testing, and account creation without detection

B. Fraud Understanding

What is the attack?

Attackers deploy **AI agents** that mimic human behavior convincingly enough to evade bot detection and behavioral biometrics. These agents "randomiz[e] mouse movements and keystrokes, mirror human delays and timings between clicks, and slow down the rate of form fills to appear human."

Darwinium's CEO notes: "AI agents are being used to trick systems into thinking they are legitimate human/trusted traffic." This represents a qualitative shift from traditional bots that were detectable by superhuman speed and unnatural behavior patterns.

Traditional mechanism:

Before GenAI, attackers used automated scripts and bots to test credentials, cards, and system vulnerabilities at scale. Bots were detectable by:

- Superhuman speed (too fast for human)
- No mouse movements or unnatural mouse paths
- Perfect form filling (no typos, no corrections)
- Consistent, linear navigation (no hesitation)
- Unusual transaction timing (e.g., 3 AM transactions)

GenAI transformation:

GenAI transforms bot interaction from **detectable scripts** to **indistinguishable human behavior**:

Dimension	Pre-GenAI	GenAI-Transformed
Behavioral mimicry	Limited, detectable	AI agents "randomiz[e] mouse movements and keystrokes, mirror human delays and timings between clicks"
Detection evasion	Static patterns	AI agents that "mimic human behavior patterns at scale"
Scale	Limited by script complexity	AI agents can operate at industrial scale
Adaptability	Fixed scripts	Adaptive behavior based on detection feedback

Why GenAI is load-bearing:

If GenAI is removed, the attacker loses the ability to:

- **Mimic human behavior** convincingly enough to evade behavioral biometrics
- **Adapt** behavior based on detection feedback
- **Operate at scale** while maintaining human-like behavior
- **Randomize** behavior patterns naturally

The attack reverts to detectable scripted bots that are **caught** by behavioral biometrics and bot detection.

HARD GenAI Removal Test result: PASS — GenAI creates a qualitatively new capability. The transformation from detectable scripted bots to indistinguishable human-mimicking AI agents is a qualitative leap.

Prerequisites:

- Access to AI agent frameworks
- Understanding of behavioral biometrics and bot detection
- Target data (credentials, card numbers, accounts)
- Infrastructure for running agents at scale

Attack Flow:

1. Attacker configures AI agents for fraud type (credential stuffing, card testing, account creation)
2. AI agents execute with human-like behavior: randomized mouse movements, natural keystroke timing, realistic navigation, appropriate form-filling speed
3. Bots attempt logins, card tests, or account creations at scale
4. Bot detection systems score the traffic as "human" (low risk)

5. Successful compromises are flagged for manual exploitation or automated fraud

Downstream payment consequence:

- Account takeover via credential stuffing at scale
- Card testing fraud (validating stolen cards)
- Mass synthetic account creation
- Bot attacks up 59% in 2025; financial services bot attacks doubled growth
- All downstream fraud enabled by successful bot operations

C. Detection Intelligence

Observable Signals:

Signal Category	Specific Signals	Detection Method	Cross-account Needed?
Device	Device fingerprint consistency	Rule-based + ML	No
Behavioral	Mouse movement naturalness	ML (behavioral biometrics)	No
Behavioral	Keystroke timing naturalness	ML (behavioral biometrics)	No
Behavioral	Form-filling patterns (typos, corrections)	ML anomaly detection	No
Behavioral	Navigation patterns (hesitation, backtracking)	ML anomaly detection	No
Behavioral	Timing patterns (human-like delays)	ML anomaly detection	No
Behavioral	Session continuity patterns	ML anomaly detection	No
Behavioral	Consistency across behavioral modalities	ML anomaly detection	No
Session	Multiple sessions from same infrastructure	Graph analytics	Yes
Network	IP/network correlation	Graph analytics	Yes
Temporal	Timing patterns (off-hours, rapid sequences)	Rule-based + ML	Yes

Evasion Considerations:

An adaptive attacker could:

- Use AI agents with **advanced behavioral mimicry**
- **Randomize** behavior patterns across sessions
- **Adapt** to detection feedback in real-time
- **Distribute** agents across different IP ranges
- Use **legitimate human behavior** as training data for agents
- **Vary** timing and patterns unpredictably

Legitimate Behavior That Could Look Similar:

- Legitimate users with variable behavior
- Users with accessibility tools
- Users in different time zones (off-hours access)
- Users with different devices (changing behavior patterns)

False-Positive Risks:

- Accessibility tools causing unusual behavior patterns
- Users with variable behavior across devices
- Legitimate high-volume users (business, corporate)

D. Simulation Intelligence

Field	Value
Simulation Type	Agentic (primary)

Field	Value
Generation Parameters	AI agent behavior parameters (mouse movements, keystroke timing, navigation patterns, form-filling speed); Bot type (credential stuffing, card testing, account creation); Scale parameters; Detection feedback adaptation
Synthetic Entities	AI agents; Synthetic traffic patterns; Behavioral profiles; Interaction sequences
Synthetic Device Attributes	Device fingerprints (spoofed or real); Browser/OS environments; IP/network context
Synthetic Session Events	Session establishment patterns; Session continuity; Multiple sessions
Synthetic Behavioral Events	Mouse movements (AI-generated, human-like); Keystroke patterns (timing, rhythm, errors); Navigation patterns (hesitation, backtracking, search); Form-filling patterns (typos, corrections)
Synthetic Transactions	Login attempts (credential stuffing); Card test attempts (card testing); Account creation attempts
Temporal Behavior	Human-like timing (delays between actions, natural speed); Session timing patterns; Off-hours patterns
Network Relationships	Multiple bots from same infrastructure; IP correlation; Agent clustering
Attack Labels	Bot type; Behavioral mimicry quality; Scale; Success rate
Variant / Mutation Parameters	Bot type; Behavioral mimicry quality; Scale; Adaptation level; Timing variance
Legitimate Counterpart	Legitimate human users; Legitimate high-volume users; Users with variable behavior
Realistic Edge Cases	Accessibility tools; Variable behavior patterns; Cross-device patterns

E. Evidence

Claim	Source	Publisher	Date	Evidence Type	Confidence
AI agents are being used to trick systems... randomizing mouse movements, mirroring human delays	Darwinium CEO	Darwinium	2025	Industry report	VERIFIED
Fraudsters now deploy agentic bots that mimic human behavior patterns at scale	Oscilar	Oscilar	2025	Industry report	VERIFIED
Fraudsters are using legitimate AI agents to bypass detection entirely, hiding in plain sight	Transmit Security	Transmit Security	2025	Industry report	VERIFIED
Bad actors spoofing AI agents to bypass malicious bot defenses	Radware	Radware	2025	Industry report	VERIFIED
Bot attacks up 59% in 2025; financial services bot attacks doubled growth	Insurance Journal	Insurance Journal	2025	News report	VERIFIED
Systems built around assumption of human initiation—automation at 3 AM is red flag	Forbes	Forbes	2025	News report	VERIFIED

F. Product Mapping

Component	Implementation
Generate	Agentic generation of AI-driven bot behavior (human-mimicking interaction); Algorithmic generation of scale parameters; Behavioral parameterization
Defend	Behavioral biometrics (mouse movements, keystroke timing, navigation patterns); Anomaly detection for behavioral patterns; Bot detection ML
Detection Approach	ML classification for behavioral anomalies; ML anomaly detection for timing patterns; Graph analytics for cross-session correlation

Component	Implementation
Mitigation	Step-up authentication for suspicious behavior; Enhanced monitoring for agent patterns; Blocking for confirmed bot activity
Evaluation	Precision (avoiding false positives on legitimate users); Recall (catching AI agents); AUC; False Positive Rate

FAMILY 5: GAN-BASED BEHAVIORAL BIOMETRICS EVASION

A. Identity

Field	Value
Attack ID	BBE-001
Attack Family	GAN-Based Behavioral Biometrics Evasion
Attack Subtypes / Variants	GAN-based keystroke injection; Robot adversarial attacks; Motion forecasting attacks; Adversarial behavioral pattern generation
Primary Lifecycle Stage	Device / Session (Stage 3)
Secondary / Cross-stage Stages	Authentication (Stage 4), Session risk assessment
Target / Actor	Behavioral biometrics systems (keystroke dynamics, mouse dynamics, touch dynamics, motion patterns)
Attack Objective	Generate behavioral patterns that match legitimate user profiles to evade behavioral biometrics detection, enabling session takeover or fraudulent activity from within an authenticated session

B. Fraud Understanding

What is the attack?

Attackers use Generative Adversarial Networks (GANs) and other GenAI techniques to generate behavioral patterns that match legitimate user profiles. This includes "Practical GAN-based Presentation Attacks on Typing Biometrics" where "GANs generate realistic keystroke patterns," as well as "Robot adversarial attacks on keystroke dynamics" and "Motion forecasting attacks on behavioral biometric authentication systems."

Unlike traditional behavioral evasion (mimicking general human behavior), GenAI-based evasion can **learn and replicate specific individual user patterns**, making it extremely difficult to detect.

Traditional mechanism:

Before GenAI, attackers attempted to mimic legitimate user behavior through:

- Statistics-based attacks (matching aggregate behavioral statistics)
- Imitation-based attacks (learning and replicating specific user patterns)
- Surrogate-based attacks (using proxy models)

These were limited in quality and detectable by modern behavioral biometrics systems.

GenAI transformation:

GenAI transforms behavioral evasion from **detectable mimicry** to **indistinguishable replication**:

Dimension	Pre-GenAI	GenAI-Transformed
Mimicry quality	Limited, detectable patterns	GANs can generate realistic keystroke patterns indistinguishable from legitimate users
Adaptation	Static	GenAI can adapt based on detection feedback
Personalization	Generic	GenAI can learn specific user patterns
Scale	Manual	Automated, scalable

Why GenAI is load-bearing:

If GenAI is removed, the attacker loses the ability to:

- **Generate indistinguishable** behavioral patterns (specifically GAN-quality keystroke and motion patterns)
- **Learn and replicate** specific individual user behavior patterns
- **Adapt** behavior based on detection feedback
- **Generate patterns** at scale that evade ML-based detection

The attack reverts to detectable, pattern-based mimicry that is **caught** by modern behavioral biometrics.

HARD GenAI Removal Test result: PASS — GenAI creates a qualitatively new capability. The transformation from detectable pattern-based mimicry to GAN-generated indistinguishable behavioral patterns is a qualitative leap.

Prerequisites:

- Access to GAN models or GenAI behavioral generators
- Training data (victim's behavioral patterns) or general human behavior training
- Understanding of behavioral biometrics systems
- Ability to inject generated behavioral patterns into interaction

Attack Flow:

1. Attacker obtains behavioral training data (from compromised device, public data, or general human behavior)
2. GAN is trained on the target's behavioral patterns (keystroke timing, mouse movements, touch gestures)
3. GAN generates realistic behavioral patterns matching the target's profile
4. Attacker injects these patterns into authentication session
5. Behavioral biometrics system scores the behavior as "legitimate" (matches target's profile)
6. Attacker conducts fraud while appearing behaviorally legitimate

Downstream payment consequence:

- Account takeover without behavioral flags
- Session hijacking without detection
- Fraud from trusted device without behavioral anomalies
- Bypass of continuous authentication
- Evasion of risk-based transaction monitoring

C. Detection Intelligence

Observable Signals:

Signal Category	Specific Signals	Detection Method	Cross-account Needed?
Behavioral	Keystroke timing naturalness	ML anomaly detection	No
Behavioral	Mouse movement naturalness	ML anomaly detection	No
Behavioral	Touch gesture naturalness	ML anomaly detection	No
Behavioral	Motion pattern naturalness	ML anomaly detection	No
Behavioral	Consistency across behavioral modalities	ML anomaly detection	No
Behavioral	Subtle deviations from user baseline	ML (behavioral biometrics)	No
Behavioral	Too-perfect mimicry (unusually consistent)	ML anomaly detection	No
Behavioral	Inconsistencies between behavioral types	ML anomaly detection	No
Session	Session anomalies (pattern changes)	ML anomaly detection	No
Transaction	Transaction anomalies (inconsistent with behavioral patterns)	ML anomaly detection	No

Evasion Considerations:

An adaptive attacker could:

- Use **GANs** to generate increasingly realistic behavioral patterns
- **Train** on specific victim's behavioral data
- **Adapt** patterns based on detection feedback

- **Combine** multiple behavioral modalities for consistency
- **Vary** patterns naturally to avoid detection

Legitimate Behavior That Could Look Similar:

- Legitimate user behavior (by definition)
- Users with variable behavior (new device, new app)
- Users with accessibility tools causing unusual behavior
- Legitimate behavioral variance

False-Positive Risks:

- Behavioral biometrics false positives on legitimate users
- Users with variable behavior across contexts
- Users with accessibility tools
- Users with physical conditions affecting behavior
- Users with new devices (new behavioral patterns)

D. Simulation Intelligence

Field	Value
Simulation Type	Agentic + Algorithmic (GAN-based generation)
Generation Parameters	Keystroke timing distributions; Mouse movement patterns; Touch gesture patterns; Motion patterns; Behavioral quality levels; Adaptation parameters
Synthetic Entities	Behavioral profiles (victim baselines); GAN-generated keystroke/mouse/touch patterns; Behavioral injection sequences; Interaction sessions
Synthetic Device Attributes	Device fingerprint (matching victim's device); Session context
Synthetic Session Events	Session establishment; Behavioral injection; Session continuity; Behavioral consistency
Synthetic Behavioral Events	Keystroke patterns (timing, rhythm, errors); Mouse movements (paths, speed, clicks); Touch gestures (swipes, pressure, timing); Motion patterns; Navigation patterns
Synthetic Transactions	Transaction sequences consistent with behavioral profile (to avoid detection)
Temporal Behavior	Timing patterns consistent with user baseline; Behavioral session patterns; Adaptation timing
Network Relationships	N/A (per-session evasion)
Attack Labels	Behavioral mimicry quality; Modality type (keystroke/mouse/touch/motion); Adaptation level; Training data source
Variant / Mutation Parameters	Mimicry quality; Modality type; Adaptation level; Training data (general vs. specific user)
Legitimate Counterpart	Legitimate user behavior (target baseline); Users with variable behavior; Users with accessibility tools
Realistic Edge Cases	Variable behavior across contexts; Accessibility tools; Physical conditions affecting behavior; New devices

E. Evidence

Claim	Source	Publisher	Date	Evidence Type	Confidence
Practical GAN-based Presentation Attacks on Typing Biometrics—GANs generate realistic keystroke patterns	IEEE	IEEE	2025	Peer-reviewed research	VERIFIED
Robot adversarial attack on keystroke dynamics based user authentication system	University of Glasgow	University of Glasgow	2025	Academic research	VERIFIED
Motion Forecasting Attacks on Behavioral Biometric Authentication Systems	ACM	ACM	2025	Peer-reviewed research	VERIFIED

Claim	Source	Publisher	Date	Evidence Type	Confidence
Criminals have found ways of circumventing behavioral biometrics by stealing credentials	LexisNexis	LexisNexis	2024	Industry report	VERIFIED
Adversarial attacks on mouse dynamics authentication: statistics-based, imitation-based, surrogate-based	arXiv	arXiv	2024	Academic research	VERIFIED
Traditional behavioral biometrics solutions struggle to keep up with modern fraud techniques	Oscilar	Oscilar	2025	Industry report	REASONABLE INFERENCE

F. Product Mapping

Component	Implementation
Generate	GAN-based generation of keystroke/mouse/touch patterns; Algorithmic parameterization of behavioral patterns; Agentic injection of generated patterns into sessions
Defend	Behavioral biometrics anomaly detection; Consistency across behavioral modalities; ML classification for GAN-generated patterns
Detection Approach	ML anomaly detection for keystroke/mouse/touch patterns; Cross-modality consistency checking; Adversarial pattern detection
Mitigation	Step-up authentication for behavioral anomalies; Enhanced monitoring for pattern inconsistencies; Multi-factor verification for high-risk sessions
Evaluation	Precision (avoiding false positives on legitimate behavior); Recall (catching GAN-generated patterns); AUC; False Acceptance Rate (FAR); False Rejection Rate (FRR)

PART 4 – FINAL COVERAGE AUDIT

Device / Session Attack Surface	Considered?	Covered By	Separate Family?	Reason
Device identity	✓	DS-01, DS-02, DS-03 (merged)	Yes	Core Device/Session surface
Device fingerprinting	✓	DS-01	Yes	Core Device/Session surface
Device reputation	✓	DS-01, DS-02	No	Detection mechanism, not attack family
Device binding	✓	THEORETICAL (DS-08)	No	Insufficient evidence for GenAI transformation
Trusted devices	✓	MERGED → DS-05 (moved to Authentication)	No	Primary attack belongs to Authentication
Device farms	✓	DS-02 (includes physical farms)	Yes	Core Device/Session surface
Emulator/virtualized environments	✓	DS-02	Yes	Core Device/Session surface
Rooted/jailbroken environments	✓	DROPPED (DS-04)	No	Cross-cutting capability, not standalone family
Automated interaction	✓	DS-04 (BOT-001)	Yes	Core Device/Session surface
Bot behavior	✓	DS-04 (BOT-001)	Yes	Core Device/Session surface
Behavioral biometrics	✓	DS-05 (BBE-001)	Yes	Core Device/Session surface
Session integrity	✓	DS-03 (RAT-001)	Yes	Core Device/Session surface
Session hijacking/manipulation	✓	MOVED → Authentication	No	Primary attack action at authentication moment
Session token abuse	✓	MOVED → Authentication	No	Primary attack action at authentication moment
Remote-access/device-control scenarios	✓	DS-03 (RAT-001)	Yes	Core Device/Session surface

Device / Session Attack Surface	Considered?	Covered By	Separate Family?	Reason
IP/network/proxy signals	✓	DROPPED (DS-10)	No	Evasion technique, not standalone family
Geolocation/device consistency	✓	DS-01	No	Detection mechanism, not attack family
Client-side integrity	✓	DS-01, DS-02, DS-03	No	Cross-cutting capability
Device enrollment/re-enrollment	✓	THEORETICAL	No	Insufficient evidence
Multi-account/device relationships	✓	MOVED → Account Creation/Cross-stage	No	Primary attack action at Account Creation

PART 5 – FINAL SUMMARY

Table 1 – FINAL DEVICE / SESSION ATTACK TAXONOMY

ID	Attack Family	Variants	GenAI Classification	Simulation Type	Primary Stage	Reason for Distinctness
DFS-001	AI-Generated Device Fingerprint Spoofing	Anti-detect browser automation; Kernel-level spoofing; AI-native anti-detection; Data wiping reset	PASS	Hybrid (Algorithmic + Agentic)	Device / Session	GenAI enables indistinguishable fingerprints; qualitative leap from template-based artifact-prone spoofing
EFF-001	AI-Driven Emulator Farm Fraud	Virtual emulator farms; Cloud phone farms; Physical device farms	PASS	Hybrid (Algorithmic + Agentic)	Device / Session	GenAI enables adaptive evasion and identity cycling; transforms detectable infrastructure to indistinguishable
RAT-001	AI-Powered Remote Access Trojan / On-Device Fraud	Automated Transfer System (ATS); Behavioral mimicry; AI-distributed RATs	PASS	Agentic + Algorithmic	Device / Session	GenAI enables autonomous execution (ATS) and behavioral mimicry; qualitative leap from human-controlled to autonomous
BOT-001	AI Agent-Powered Automated Bot Interaction	Human-mimicking credential stuffing; Human-mimicking card testing; Autonomous agent fraud	PASS	Agentic	Device / Session	GenAI enables indistinguishable human behavior; qualitative leap from detectable scripts to human-mimicking agents
BBE-001	GAN-Based Behavioral Biometrics Evasion	GAN-based keystroke injection; Robot adversarial attacks; Motion forecasting attacks	PASS	Agentic + Algorithmic	Device / Session	GenAI (GANs) enables indistinguishable behavioral patterns; qualitative leap from detectable mimicry to GAN-generated patterns

Total: 5 final Device / Session attack families

Table 2 – ATTACK × OBSERVABLE SIGNAL

Attack Family	Device	Session	Behavioral	Transaction	Network	Temporal
DFS-001: Device Fingerprint Spoofing	Canvas/WebGL/audio artifacts; Font/timezone consistency; Hardware attributes; Multiple accounts same fingerprint	Session token consistency; Fingerprint cycling patterns	Mouse movements; Keystroke patterns; Navigation patterns	Transaction patterns (downstream)	Account-to-account similarity; Fingerprint pattern clustering	Fingerprint cycling frequency; Session timing
EFF-001: Emulator Farm Fraud	Emulator/VM flags; Cloud/VPS IP ranges; Identical fingerprints across accounts	Account creation velocity; Multiple accounts same IP/device	Automated patterns; Behavioral consistency	Mass account creation; Dropper transactions	IP/network correlation; Fraud ring clustering	Off-hours activity; Identical account creation patterns

Attack Family	Device	Session	Behavioral	Transaction	Network	Temporal
RAT-001: RAT/On-Device Fraud	Device fingerprint (consistent); Suspicious app installations; Screen recording indicators	Session anomalies; ATS execution patterns	Behavioral anomalies; Unusual navigation; Timing anomalies	Unusual amounts/beneficiaries; Rapid sequences; ATS execution	Beneficiary network correlations	Transaction timing; Behavior pattern timing; Time to fraud
BOT-001: Automated Bot Interaction	Device fingerprint consistency	Multiple sessions same infrastructure	Mouse movement naturalness; Keystroke timing; Form-filling patterns	Login/card test/account creation attempts	IP/network correlation	Off-hours activity; Rapid sequences
BBE-001: Behavioral Biometrics Evasion	N/A (consistent)	Session anomalies	Keystroke/mouse/touch patterns; Cross-modality consistency	Transaction consistency	N/A	Timing patterns; Session patterns

Table 3 – ATTACK × SIMULATION

Attack Family	Synthetic Entities	Device/Session Events	Behavioral Events	Transactions	Network	Agentic/Algorithmic/Hybrid
DFS-001	Device profiles; Browser/session environments; Device histories	Session establishment; Cookie/token generation	Mouse movements; Keystroke patterns; Navigation; Form-filling	N/A	Multiple accounts sharing fingerprint	Hybrid
EFF-001	Device profiles; Emulator instances; Account creation attempts	Multiple session establishment; Account creation sequences	AI-generated human-like behavior; Evasion patterns	Account creation sequences	Multiple accounts per device; IP correlation; Fraud ring clustering	Hybrid
RAT-001	Victim personas; RAT-infected devices; ATS sequences	Normal session; Session takeover; ATS execution	Victim baseline; Attacker mimicry; ATS behavior	Normal transactions; ATS-executed fraud; Beneficiary additions	Mule network connections	Agentic + Algorithmic
BOT-001	AI agents; Synthetic traffic; Behavioral profiles	Session establishment patterns; Multiple sessions	Mouse movements; Keystroke patterns; Navigation; Form-filling	Login attempts; Card tests; Account creation attempts	Multiple bots same infrastructure; IP correlation	Agentic
BBE-001	Behavioral profiles; GAN-generated patterns; Interaction sessions	Session establishment; Behavioral injection	Keystroke/mouse/touch/motion patterns	Transaction sequences	N/A	Agentic + Algorithmic

Table 4 – TAXONOMY DECISIONS

Candidate	Decision	Reason
DS-01: Device Fingerprint Spoofing	KEEP	Core Device/Session family. GenAI enables indistinguishable fingerprints.
DS-02: Emulator Farm Fraud	KEEP	Core Device/Session family. GenAI enables adaptive evasion.
DS-03: Physical Device Farm Fraud	MERGE → DS-02	Same mechanism as emulator farms; physical vs. virtual is implementation variant.

Candidate	Decision	Reason
DS-04: Rooted/Jailbroken Device Abuse	DROP	Cross-cutting capability, not standalone family; GenAI is amplification, not transformation.
DS-05: Session Hijacking	MOVE → Authentication	Primary attack action at authentication moment; GenAI is amplification, not transformation.
DS-06: RAT/On-Device Fraud	KEEP	Core Device/Session family. GenAI enables ATS and behavioral mimicry—qualitative leaps.
DS-07: Automated Bot Interaction	KEEP	Core Device/Session family. GenAI enables human-mimicking behavior—qualitative leap.
DS-08: Device Binding Bypass	THEORETICAL	Insufficient real-world evidence; attack works fully without GenAI.
DS-09: Trusted Device Abuse	MERGE → DS-05	Variant of session/token abuse; move with DS-05 to Authentication.
DS-10: IP/Proxy/VPN Evasion	DROP	Evasion technique, not standalone family; GenAI is incidental.
DS-11: Behavioral Biometrics Evasion	KEEP	Core Device/Session family. GAN-based attacks are a qualitative leap.
DS-12: Multi-Accounting	MOVE → Account Creation/Cross-stage	Primary attack action is creating accounts; Device/Session component is DS-01.
DS-13: Data Wiping	MERGE → DS-01	Data wiping is a technique to reset fingerprints; not a distinct family.

FINAL SUMMARY

Metric	Value
Total final attack families	5
Total variants	14 (across 5 families)
Important candidates merged	DS-03 (Physical Device Farm → DS-02); DS-09 (Trusted Device Abuse → DS-05); DS-13 (Data Wiping → DS-01)
Important candidates moved	DS-05 (Session Hijacking → Authentication); DS-09 (Trusted Device Abuse → Authentication); DS-12 (Multi-Accounting → Account Creation/Cross-stage)
Important candidates dropped	DS-04 (Rooted/Jailbroken); DS-10 (IP/Proxy/VPN)
Theoretical/future threats	DS-08 (Device Binding Bypass) — plausible but insufficient real-world evidence
Remaining research gaps	Live deployment evidence for GAN-based behavioral evasion in financial fraud; Real-world frequency of AI-agent-powered bot interaction; Specific detection evasion effectiveness

Final Note on Taxonomy Consistency

The Device/Session taxonomy now has **5 families**, consistent with:

- **Identity/KYC** = 5 families
- **Account Creation/Onboarding** = 3 families

The number 5 emerged naturally from the evidence and the GenAI Removal Test—not forced. Each of the 5 families represents a **distinct attack mechanism** with **load-bearing GenAI transformation**, **strong evidence**, and **simulation feasibility**.

The **unifying theme** across all 5 Device/Session families is the **shift from "appearing different" to "appearing identical"** —fraudsters no longer try to evade detection by being different; they use GenAI to appear **indistinguishable** from legitimate users and devices.